STA642

Probability Distribution

Short Notes for Final Term

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Note:

PMF; Probability Mass Function
PDF; Probability Density Function
CDF; Cumulative Distribution Function
MGF; Moment Generating Function
CGF; Cumulant Generating Function

Formal Definition of Independence in the case of Discrete Random Variables

Let the random variables X_1 and X_2 have the joint pmf $p(x_1,x_2)$ and the marginal pmfs $p_1(x_1)$ and $p_2(x_2)$ respectively.

The random variables X_1 and X_2 are said to be independent **if and only if**

$$p(x_1, x_2) = p_1(x_1)\, p_1(x_2).$$

Theorem:

If X and Y are random variables with joint distribution function F and marginal distribution functions F_X and F_Y, then X and Y are independent if and only if

$$F(x, y) \equiv F_X(x) F_Y(y)$$

For all $(x,y) \in R^2$.

Proof:

If X and Y are independent, then

$$F(x,y)=P(X\le x,Y\le y)=P(X\le x)P(Y\le y)=F_X(x)F_Y(y).$$

Theorem:

Suppose X_1 and X_2 are independent and that $E(u(X_1))$ and $E(v(X_2))$ exist.

Then

$$E\left[u(X_1)v(X_2)\right]=E\left[u(X_1)\right]E\left[v(X_2)\right].$$

Theorem:

Suppose the joint mgf, $M(t_1,t_2)$, exists for the random variables X_1 and X_2.

Then, X_1 and X_2 are independent if and only if

$$M(t_1,t_2)=M(t_1,0)M(0,t_2);$$

That is, the joint mgf is identically equal to the product of the marginal mgfs.

PMF of the Discrete Uniform Distribution

If x is assuming values 1,2,3,…,n, then the probability mass function (PMF) of the discrete uniform distribution is given by

$$p(x)=\begin{cases}\dfrac{1}{n}, & x=1,2,3,...,n\\ 0 & elsewhere\end{cases}$$

CDF of the Discrete Uniform Distribution

$$p(X=x)=\begin{cases}\dfrac{1}{b-a+1}, & x=a,a+1,a+2,...,b-1,b\\ 0 & elsewhere\end{cases}$$

Mean of the Discrete Uniform Distribution

$$\mu=E(X)=\frac{n+1}{2}$$

Variance of the Discrete Uniform Distribution

$$\sigma=E(X^2)-\left(E(X)\right)^2=\sqrt{\frac{n^2-1}{12}}$$

MGF of the Discrete Uniform distribution

Moment Generating Function:

$$M(t)=E\left(e^{tx}\right)=\sum_{x=1}^{N}\left(e^{tx}\frac{1}{n}\right)=\frac{1}{n}\sum_{x=1}^{n}\left(e^{t}\right)^{x}=\frac{e^{t}-e^{(n+1)t}}{n\left(1-e^{t}\right)}$$

PMF of the Binomial distribution

Let the random variable X denote the number of successes in *n* trials --- when a binomial experiment is performed.

Then the PMF of X is given by

$$p(x)=\begin{cases}\binom{n}{x}p^{x}(1-p)^{n-x} & x=0,1,2,\ldots,n\\ 0 & elsewhere.\end{cases}$$

where *p* is the probability of success in each trial and *n* is the number of trials.

Binomial Theorem

$$(q+p)^{n}=\sum_{x=0}^{n}\binom{n}{x}p^{x}q^{n-x}$$

Mean of the Binomial Theorem

$$\mu=np$$

Variance of the Binomial Theorem

$$\sigma^{2}=np(1-q)=npq$$

MGF of the Binomial distribution

The mgf of a binomial distribution is easily obtained as follows:

$$\begin{aligned}M(t)&=\sum_{x}e^{tx}p(x)=\sum_{x=0}^{n}e^{tx}\binom{n}{x}p^{x}(1-p)^{n-x}\\&=\sum_{x=0}^{n}\binom{n}{x}\left(e^{t}p\right)^{x}(1-p)^{n-x}\\&=\left[(1-p)+pe^{t}\right]^{n}\\M(t)&=\left(1-p+pe^{t}\right)^{n}\end{aligned}$$

for all real values of *t*.

So, in case of m binomial random variables X_i, $i = 1, 2, ..., m$,
with $p_1 = p_2 = p_3 ..... p_m = p$ (i.e. a constant)
the mgf of X_i is $M_{X_i}(t) = \left(1 - p + pe^t\right)^{n_i}$

PMF of the Negative Binomial Distribution

In a sequence of such trials, let the random variable X denote the number of the trial at which the ***r*th success** occurs, where r is a fixed integer.

$$P(X = x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, \; x = r, r+1, r+2, ...$$

and we say that X has a negative binomial (*r*, *p*) distribution.

Mean of the Negative Binomial Distribution

$$E(X) = \mu = \frac{r(1-p)}{p} = \frac{rq}{p}$$

Variance of the Negative Binomial Distribution

$$\text{Var}(X) = \sigma^2 = \frac{r(1-p)}{p^2} = \frac{rq}{p^2}$$

MGF of the Negative Binomial Distribution

$$M(t) = \left(\frac{1-p}{1-pe^t}\right)^r \quad for \;\; t < -\log p$$

A Special Case of the Negative Binomial Distribution: Geometric Distribution

In a sequence of independent Bernoulli (p) trials, let the random variable X denote the number of the trial at which the ***1st* success** occurs.

PMF of the Geometric Distribution

$$P(X = x) = \binom{x-1}{0} p^1 (1-p)^{x-1}$$
$$= (1-p)^{x-1} p, \quad x = 1, 2, 3, ...$$

Mean of the Geometric Distribution

$$\mu = E(X) = \frac{1}{p}$$

Variance of the Geometric Distribution

$$\sigma^2 = Var(X) = \frac{1-p}{p^2} = \frac{q}{p^2}$$

Multinomial Distribution

The binomial distribution is generalized to be the multinomial distribution as follows:

Let p_i be the probability that the outcomes is an element of C_i and let p_i remain constant throughout the n independent repetitions, i= 1,2,…,k.

PMF of the Multinomial Distribution

$$f(x_1, x_2, ..., x_k; n, p_1, p_2, ..., p_k) = \begin{cases} \dfrac{n!}{x_1! x_2! \cdots x_k!} p_1^{x_1} \times p_2^{x_2} \cdots \times p_k^{x_k}, & \text{when } \sum_{i=1}^{k} x_i = n \\ 0 & \text{otherwise} \end{cases}$$

for non-negative integers $x_1, x_2, ..., x_k$.

Mean of the Multinomial Distribution

$$\mu = E(X_i) = np_i$$

Variance of the Multinomial Distribution

$$\sigma^2 = \text{Var}(X_i) = np_i(1-p_i) = np_iq_i$$

MGF of the Multinomial Distribution

$$M_{\underline{X}}(t) = M_{\underline{X}}(t_1, t_2, \ldots, t_k) = E\left(e^{\sum_{i=1}^{k} t_i X_i}\right) = \left(\sum_{i=1}^{k} p_i e^{t_i}\right)^n$$

CGF of the Multinomial Distribution

Cumulant Generating Function:

$$K_{\underline{X}}(t) = K_{\underline{X}}(t_1, t_2, \ldots, t_k) = \log\left[M_{\underline{X}}(t_1, t_2, \ldots, t_k)\right] = n \log\left(\sum_{i=1}^{k} p_i e^{t_i}\right)$$

MGF of the Trinomial Distribution

$$M_{\underline{X}}(t) = M_{X_1, X_2, X_3}(t_1, t_2, t_3) = E\left(e^{\sum_{i=1}^{3} t_i X_i}\right)$$

$$M_{\underline{X}}(t) = \left(\sum_{i=1}^{3} p_i e^{t_i}\right)^2 = \left(p_1 e^{t_1} + p_2 e^{t_2} + p_3 e^{t_3}\right)^2$$

PMF of the Hypergeometric Distribution

A random variable X follows the hypergeometric distribution if its probability mass function (pmf) is given by

$$P(X=x)=\frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}},\ x=0,1,2,...,K\ if\ K<n\ \ \&\ \ x=0,1,2,...,n\ if\ n<K$$

where,

- N is the population size,

- *K* is the number of success in the population at the beginning of the experiment,

- *n* is the number of draws (i.e. quantity drawn in each trial),

- *k* is the number of observed successes

Mean of the Hypergeometric Distribution

$$\text{Mean}=\mu=E(X)=n\frac{K}{N}$$

Variance of the Hypergeometric Distribution

$$\text{Variance}=\sigma^2=n\frac{K}{N}\frac{(N-K)}{N}\frac{N-n}{N-1}$$

PMF of the Poisson Distribution

$$p(x)=\begin{cases}\dfrac{e^{-\lambda}\lambda^x}{x!} & x=0,1,2,...,\ \ \lambda>0\\ 0 & elsewhere,\end{cases}$$

Mean and Variance of the Poisson Distribution

$$\text{Mean}=\text{Variance}=\lambda>0$$
$$\mu=\sigma^2=\lambda$$

Coefficient of Variation of the Poisson Distribution

$$COV=\frac{\sigma}{\mu}=\frac{\sigma}{\mu}\times100\%=\frac{\sqrt{\lambda}}{\lambda}\times100\%=\frac{1}{\sqrt{\lambda}}\times100\%$$

MGF of the Poisson Distribution

$$M(t)=\sum_x e^{tx}p(x)=\sum_{x=0}^{\infty}e^{tx}\frac{m^x e^{-m}}{x!}=e^{-m}\sum_{x=0}^{\infty}\frac{\left(me^t\right)^x}{x!}=e^{-m}e^{me^t}=e^{m\left(e^t-1\right)}$$

PDF of the Continuous Uniform Distribution

$$f(x)=\begin{cases}\dfrac{1}{b-a} & for\ a\le x\le b\\ 0 & for\ x<a\ or\ x>b\end{cases}$$

CDF of the Continuous Uniform Distribution

$$F(x)=\begin{cases}0 & for\ x<a\\ \dfrac{x-a}{b-a} & for\ a\le x\le b\\ 1 & for\ x>b\end{cases}$$

Mean of the Continuous Uniform distribution

$$\mu=E(X)=\frac{(a+b)}{2}$$

Variance of the Continuous Uniform distribution

$$\sigma^2=\mathrm{Var}(X)=\frac{(b-a)^2}{12}$$

PDF of the Exponential Distribution

$$f(x;\lambda)=\begin{cases}\lambda e^{-\lambda x} & x\ge 0,\ \lambda>0\\ 0 & x<0\end{cases}$$

Let $\theta=\dfrac{1}{\lambda}$ then, mean $=\theta$, and

$$f(x;\theta)=\begin{cases}\dfrac{1}{\theta}e^{-\frac{x}{\theta}} & x\ge 0,\ \theta>0\\ 0 & x<0\end{cases}$$

CDF of the Exponential Distribution

$$F(x;\lambda)=\begin{cases}1-e^{-\lambda x} & x\ge 0,\ \lambda>0\\ 0 & x<0\end{cases}$$

or

$$F(x;\theta)=\begin{cases}1-e^{-\frac{x}{\theta}} & x\geq 0,\ \theta>0\\ 0 & x<0\end{cases}$$

Mean of the Exponential Distribution

$$\mu=E(X)=\frac{1}{\lambda}$$

Variance and Standard Deviation of the Exponential Distribution

$$Var(X)=\sigma^2=\frac{1}{\lambda^2}$$

$$S.D(X)=\sigma=\frac{1}{\lambda}$$

Coefficient of Variation of the Exponential Distribution

$$CV=\frac{\sigma}{\mu}=\frac{1/\lambda}{1/\lambda}=\frac{\lambda}{\lambda}=1$$

MGF of the Exponential Distribution

$$M_0(t)=E\left[e^{tX}\right]=\frac{\lambda}{\lambda-t}$$

PDF of the Gamma Distribution

$$f(x)=\begin{cases}\frac{1}{\Gamma(\alpha)\beta^{\alpha}}x^{\alpha-1}e^{-x/\beta} & 0<x<\infty,\ \alpha>\ 0,\ \beta>0\\ 0 & \text{elsewhere}\end{cases}$$

Gamma Function

$$\Gamma(\alpha)=\int_0^{\infty}y^{\alpha-1}e^{-y}dy$$

CDF of the Gamma Distribution

$$F(x)=\begin{cases}\frac{1}{\Gamma(\alpha)}\gamma\left(\alpha,\frac{x}{\beta}\right) & \alpha>0,\beta>0\\ 0 & \text{elsewhere}\end{cases}$$

where this $\gamma\left(\alpha,\frac{x}{\beta}\right)$ is the incomplete gamma function.

Incomplete Gamma Function

In mathematics, the **upper** and **lower incomplete gamma functions** are types of special functions which arise as solutions to various mathematical problems such as certain integrals.

The upper incomplete gamma function is defined as: $\Gamma(\alpha, y) = \int_{y}^{\infty} t^{\alpha-1} e^{-t} dt$

whereas the lower incomplete gamma function is defined as: $\gamma(\alpha, y) = \int_{0}^{y} t^{\alpha-1} e^{-t} dt$

It is interesting to note that $\gamma(\alpha, y) + \Gamma(\alpha, y) = \Gamma(\alpha)$.

Mean of the Gamma Distribution

$$\mu_X = E(X) = \alpha\beta$$

Variance of the Gamma Distribution

$$Var(X) = \sigma_X^2 = \alpha\beta^2$$

Mode of the Gamma Distribution

$$\text{Mode} = (\alpha - 1)\beta$$

MGF of the Gamma Distribution

$$M(t) = (1 - \beta t)^{-\alpha} \quad for\ t < \frac{1}{\beta}$$

PDF of the Beta Distribution

$$f(x; \alpha, \beta) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}, \qquad 0 \le x \le 1; \alpha, \beta > 0$$

$$\text{where, } B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}$$

CDF of the Beta Distribution

The formula for the cumulative distribution function of the beta distribution is also called the **incomplete β function ratio** (commonly denoted by I_x) and is defined as

$$F(x) = I_x(\alpha, \beta) = \frac{\int_{0}^{x} t^{\alpha-1} (1-t)^{\beta-1}}{B(\alpha, \beta)}, \qquad 0 \le x \le 1; \alpha, \beta > 0$$

Mean of the Beta Distribution

$$\mu = E(X) = \frac{\alpha}{(\alpha+\beta)}$$

Variance of the Beta Distribution

$$Var(X) = \sigma^2 = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

Mode of the Beta Distribution

$$\text{Mode} = \frac{\alpha-1}{\alpha+\beta-2}$$

PDF of the Normal Distribution

A random variable X is said to have a normal distribution if its pdf is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \text{for } -\infty < x < \infty.$$

μ is the mean or expectation, $\sigma > 0$ is the standard deviation, and σ^2 is the variance.

CDF of the Normal Distribution

$$F(x) = \frac{1}{2}\left[1 + \text{erf}\left(\frac{x-\mu}{\sigma\sqrt{2}}\right)\right] \quad \text{for } -\infty < x < \infty$$

erf (z) is the "error function" defined by $\text{erf}(z) \equiv \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt$

Mean of the Normal Distribution

$$Mean = E(X) = \mu$$

Median of the Normal Distribution

$$Median = \mu$$

Mode of the Normal Distribution

$$Mode = \mu$$

Mean of the Normal Distribution

$$Var(X) = \sigma^2$$

Mean of the Normal Distribution

$$M(t) = E\left(e^{tX}\right) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}, \quad \text{for } -\infty < t < \infty$$

Cumulant Generating Function of the Normal distribution

$$K(t) = \ln E\left(e^{tX}\right) = \ln e^{\mu t + \frac{1}{2}\sigma^2 t^2} = \mu t + \frac{\sigma^2 t^2}{2}.\ln e = \mu t + \frac{\sigma^2 t^2}{2}(1) = \mu t + \frac{\sigma^2 t^2}{2}$$

The cumulants of the normal distribution with expected value μ and variance σ^2 are

$K_1 = \mu, \quad K_2 = \sigma^2$
and
$K_3 = K_4 = \ldots = 0.$

Standard Normal Distribution

The standard normal distribution is a **special** case of the normal distribution.

It is the distribution that occurs when a normal random variable has a **mean** of zero and a **standard deviation** of one.

PDF of the Standard Normal Distribution

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{\left(-\frac{z^2}{2}\right)}, \quad -\infty < z < \infty$$

Mean of the Standard Normal Distribution

$$\mu = E(Z) = \int_{-\infty}^{\infty} z f_Z(z)\, dz = 0$$

Variance of the Standard Normal Distribution

$$Var(Z) = \sigma^2 = 1$$

Semi-Inter Quartile Range or Quartile Deviation

$$Q.D = \frac{Q_3 - Q_1}{2}$$

Some Relationships for Standard Normal Distribution

a) $Q/\sigma = 0.6745 \quad or \quad Q = 0.6745\sigma$
b) $Q_1 = \mu - 0.6745\sigma \quad and \quad Q_3 = \mu + 0.6745\sigma$
c) $x = \mu + \sigma \quad and \quad x = \mu - \sigma$

The normal curve has Points of Inflection which are equidistant from the mean.

Two points of inflection of normal curve are

$$\left[\mu-\sigma,\frac{1}{\sigma\sqrt{2\pi e}}\right] and \left[\mu+\sigma,\frac{1}{\sigma\sqrt{2\pi e}}\right].$$

In other words, the points of inflection occur on the right and on the left of the mean at a distance equal to standard deviation and thus the graph of the normal curve are bell-shaped.

Theorem:

For the normal distribution, the **odd** order moments about the mean are all zero.

$$\mu_1=\mu_3=\mu_5=\cdots=0$$

For the normal distribution, the **odd** order moments about the mean are given by

$$\mu_{2n}=(2n-1)(2n-3)\ldots 5.3.1\,\sigma^{2n}$$

PDF of the Bivariate Normal Distribution

A pair of random variables X_1 and X_2 has a bivariate normal distribution and they are referred to as **jointly normally distributed** random variables **if and only if** their joint probability density is given by

$$f(x_1,x_2)=\frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}}e^{\left\{-\frac{1}{2(1-\rho^2)}\left(\frac{(x_1-\mu_1)^2}{\sigma_1^2}-2\rho\left(\frac{x_1-\mu_1}{\sigma_1}\right)\left(\frac{x_2-\mu_2}{\sigma_2}\right)+\frac{(x_2-\mu_2)^2}{\sigma_2^2}\right)\right\}}$$

$$\text{for } -\infty<x_1<\infty \text{ and } -\infty<x_2<\infty,$$

$$\text{where } \mu_1\in\mathbb{R},\mu_2\in\mathbb{R},\sigma_1>0,\sigma_2>0, \text{ and } -1\le\rho\le 1.$$

This is referred to as the $N\left(\mu_x,\ \mu_y,\ \sigma_x^2,\sigma_y^2,\rho\right)$ distribution.

MGF of the Bivariate Normal Distribution

$$M_{\underline{X}}(t)=M_{\underline{X}}(t_1,t_2)=E\left(e^{\sum_{i=1}^{2}t_iX_i}\right)$$

$$M_{\underline{X}}(t)=e^{\mu_1t_1+\mu_2t_2+\frac{1}{2}\left(\sigma_1^2t_1^2+2\rho\sigma_1+\sigma_2t_1t_2+\sigma_2^2t_2^2\right)},\ -\infty<t_1<\infty\ ;\ -\infty<t_2<\infty$$

Marginal of the Bivariate Normal Distribution

For any bivariate distribution $f(x,y)$, the marginal distribution of X is given by

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)\,dy, \quad -\infty < x < \infty$$

$$f_X(x) = \int_{-\infty}^{\infty} \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} \mathrm{e}^{-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_x)^2}{\sigma_x^2}+\frac{(y-\mu_y)^2}{\sigma_y^2}-\frac{2\rho(x-\mu_x)(y-\mu_y)}{\sigma_x\sigma_y}\right]} dy$$

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_x} e^{-\frac{1}{2}\left(\frac{x-\mu_x}{\sigma_1}\right)^2}, \quad -\infty < x < \infty$$

i.e. the normal distribution with μ_x and Variance σ_x^2.

$$i.e.\ X \sim N\left(\mu_x, \sigma_x^2\right) \text{ with } \mathrm{E}(X) = \mu_x,\ \mathrm{Var}(X) = \sigma_x^2,$$

Similarly, $Y \sim N\left(\mu_y, \sigma_y^2\right)$ with $\mathrm{E}(Y) = \mu_y,\ \mathrm{Var}(Y) = \sigma_y^2.$

Theorem:

If X and Y have a bivariate normal distribution with correlation coefficient ρ. Then X and Y are independent **if and only if** $\rho = 0$.

Definition of Independence:

Two random variables X and Y are independent, statistically independent or stochastically independent if

$$f_{XY}(x, y) = f_X(x)\, f_Y(y)$$

i.e. if the joint density $f_{XY}(x, y)$ is equal to the product of the marginal density $f_X(x)$ and the marginal density $f_Y(y)$.

Conditional distributions in the case of Bivariate Normal Distribution

The conditional distribution of Y given x can be expressed as follows:

$$N\left(\mu_y + \rho\frac{\sigma_y}{\sigma_x}(x-\mu_x),\ \sigma_y^2\left(1-\rho^2\right)\right) = N\left(\tilde{\mu}_y,\ \tilde{\sigma}_y^2\right)$$